Example 2.11 The area of the circular region is 2464 cm^2. Find its radius and diameter. (use $\pi = \frac{22}{7}$)

Solution

Given that the area of the circular region $= 2464\ cm^2$

$$\pi r^2 = 2464$$

$$\frac{22}{7} \times r^2 = 2464$$

$$r^2 = 2464 \times \frac{7}{22}$$

$$r^2 = 112 \times 7 = 784$$

$$r^2 = 2 \times 2 \times 2 \times 2 \times 7 \times 7$$

$$= 4 \times 4 \times 7 \times 7$$

$$= 4^2 \times 7^2$$

$$r^2 = (4 \times 7)^2 \qquad [r \times r = (4 \times 7) \times (4 \times 7)]$$

$$r = 4 \times 7$$

$$= 28\ cm$$

Diameter $(d) = 2 \times r = 2 \times 28 = 56\ cm.$

2	784
2	392
2	196
2	98
7	49
	7

Example 2.12 A gardener walks around a circular park of distance 154 m. If he wants to level the park at the rate of ₹25 per $sq.m$, how much amount will he need? (use $\pi = \frac{22}{7}$)

Solution

The distance covered by the man is nothing but the circumference of the circle.

Given that the distance covered $= 154\ m$

Therefore circumference of the circle $= 154\ m$

That is, $2\pi r = 154$

$$2 \times \frac{22}{7} \times r = 154$$

$$r = 154 \times \frac{7}{44}$$

$$r = 3.5 \times 7$$

$$= 24.5$$

Area of the park $= \pi r^2\ sq.units$

$$= \frac{22}{7} \times 24.5 \times 24.5$$

$$= 22 \times 3.5 \times 24.5$$

$$= 1886.5\ m^2$$

Cost of levelling the park per sq.m $=$ ₹ 25.

Cost of levelling the park of $1886.5 m^2 = 1886.5 \times 25 =$ ₹ 47,162.50

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